

Module 10: Epigenetics - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. How does epigenetic regulation differ from a mutation?
2. A. Both permanently change the DNA sequence.
3. B. Epigenetics deletes entire genes.
4. C. Epigenetics changes gene activity without changing the DNA sequence.
5. D. Mutations never affect any trait.
6. Answer: C

7. Why: Epigenetic marks such as methylation change how genes are expressed, not the underlying sequence. A mutation changes the sequence itself.

8. DNA methylation is usually associated with:
9. A. reduced transcription of a gene
10. B. faster DNA replication
11. C. deletion of a chromosome
12. D. building proteins directly
13. Answer: A

14. Why: Methylation typically silences or reduces transcription by making the DNA less accessible.

15. When chromatin is packed tightly, genes in that region are generally:
16. A. copied much faster
17. B. more likely to mutate
18. C. more accessible to be read
19. D. less accessible and less expressed
20. Answer: D

21. Why: Compact chromatin blocks access to genes, so they are expressed less. Open chromatin allows transcription.

22. Two genetically identical twins come to differ in a trait over their lifetimes. A good explanation is that:

23. A. their DNA sequences must actually be different

24. B. environment and epigenetic changes altered their gene expression

25. C. one twin is not truly human

26. D. epigenetic change is impossible

27. Answer: B

28. Why: Identical DNA can still show different expression because environment-driven epigenetic changes turn genes up or down.